Lecture 8. Classification I

Supervised learning, naive Bayes, LDA and QDA, logistic regression and cross-entropy loss

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8.1 Supervised learning

Motivation. Classification (phishing? churn? click?) is prediction with a categorical outcome.

Supervised learning trains on data with known outcomes: prediction (numeric) and **classification** (categorical). Clustering (no response) is **unsupervised**.

Prediction. LOESS of `volume` on `avgtemp`: the response "supervises" the fit.

```
RailTrail %>%
  mutate(loess.fit = loess(volume ~ avgtemp, data = .)$fitted) %>%
  select(volume, loess.fit, avgtemp) %>% head(4)
```

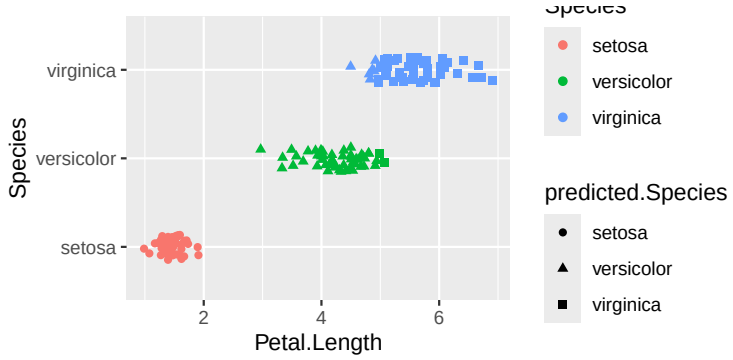
```
##   volume loess.fit avgtemp
## 1    501  469.4541    66.5
## 2    419  443.2781    61.0
## 3    397  462.1633    63.0
## 4    385  400.6197    78.0
```

Classification. Petal length almost separates the `iris` species; a crude rule (setosa < 2, versicolor 2–5, virginica > 5):

```
iris.crude <- iris %>%
  mutate(predicted.Species = ifelse(Petal.Length < 2, "setosa",
                                    ifelse(Petal.Length < 5, "versicolor", "virginica")))
iris.crude %>% count(Species, predicted.Species)
```

```
##      Species predicted.Species  n
## 1     setosa            setosa 50
## 2 versicolor        versicolor 48
## 3 versicolor         virginica  2
## 4  virginica        versicolor  6
## 5  virginica         virginica 44
```

```
ggplot(iris.crude, aes(Petal.Length, Species, colour = Species, shape = predicted.
    Species)) +
  geom_jitter(height = 0.15)
```



Two variables can be eyeballed; four or more need models or algorithms.

8.2 Naive Bayes

Motivation. Bayes' theorem turns easy $P(x \mid y)$ into the wanted $P(y \mid x)$.

Example. 1994 US Census (32 561 adults; 24% earn over \$50 000). With $y = 1$ for high earners and $x = 1$ for male, $P(y = 1) = 0.241$, $P(\text{male}) = 0.670$ and $P(\text{male} \mid y = 1) = 0.845$, so by Bayes'

theorem

$$P(y=1 \mid \text{male}) = \frac{P(\text{male} \mid y=1)\,P(y=1)}{P(\text{male})} = \frac{0.845 \times 0.241}{0.670} \approx 0.30.$$

Assign x^* to $y=1$ when $P(y=1 \mid x^*) > P(y=0 \mid x^*)$. Naive Bayes models $P(x_1,\dots,x_p \mid y)$ assuming **conditional independence** given the class:

$$P(x_1,\dots,x_p \mid y) = \prod_{j=1}^{p} P(x_j \mid y),$$

so for classes $k = 1,\dots,K$,

$$P(y=i \mid x_1,\dots,x_p) = \frac{\prod_j P(x_j \mid y=i)\; P(y=i)}{\sum_{k=1}^{K} \prod_j P(x_j \mid y=k)\; P(y=k)}.$$

Numeric predictors usually get a normal $P(x_j \mid y)$. Naive Bayes often works well. Data: 10 000 credit-card customers in `ISLR2::Default` (did they default?), split 80/20:

```
set.seed(364)
idx <- sample(nrow(Default), 0.8 * nrow(Default))
train <- Default[idx, ]; test <- Default[-idx, ]
mean(train$default == "Yes")
```

```
## [1] 0.03125
```

```
accuracy <- function(pred, truth) mean(pred == truth)
mod_nb <- naiveBayes(default ~ student + balance + income, data = train)
accuracy(predict(mod_nb, test), test$default)
```

```
## [1] 0.9625
```

Only 3% default, so always "No" is 97% accurate: accuracy is poor for unbalanced classes (Lecture 9) [Supplement].

8.3 Discriminant analysis

Motivation. With normal classes and a common covariance, the Bayes rule is linear: LDA (Fisher, 1936).

One predictor. $f_k(x) = \phi(x;\mu_k,\sigma^2)$, prior $\pi_k = P(y=k)$. With $\hat\pi_k = n_k/n$, $\hat\mu_k$, $\hat\sigma^2$, assign x to the largest **discriminant function**

$$\delta_k(x) = x\,\frac{\hat\mu_k}{\hat\sigma^2} - \frac{\hat\mu_k^2}{2\hat\sigma^2} + \log\hat\pi_k.$$

Proposition 8.1 (LDA discriminant)

Under the model above, $P(y=k \mid x)$ is largest for the k maximising $\delta_k(x)$. With two classes and $\pi_0 = \pi_1$, the boundary is $x = (\mu_0+\mu_1)/2$.

Proof [Supplement]. $P(y=k \mid x) \propto \pi_k f_k(x)$, with the same denominator for all k. Taking logs, $\log\pi_k - \frac{(x-\mu_k)^2}{2\sigma^2} + \text{const} = \log\pi_k + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} - \frac{x^2}{2\sigma^2} + \text{const}$; the last two terms are common to all classes. With equal priors, $\delta_1(x) > \delta_0(x)$ iff $x(\mu_1-\mu_0)/\sigma^2 > (\mu_1^2-\mu_0^2)/(2\sigma^2)$, i.e. (for $\mu_1 > \mu_0$) iff $x > (\mu_0+\mu_1)/2$. ∎

Several predictors. Classes $N(\mu_k,\Sigma)$:

$$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\Big\{-\frac{1}{2}(x-\mu_k)^{\mathsf T}\Sigma^{-1}(x-\mu_k)\Big\}.$$

Dropping common terms of the log,

$$\delta_k(x) = x^\top \hat{\Sigma}^{-1}\hat{\mu}_k - \frac{1}{2}\hat{\mu}_k^\top \hat{\Sigma}^{-1}\hat{\mu}_k + \log \hat{\pi}_k,$$

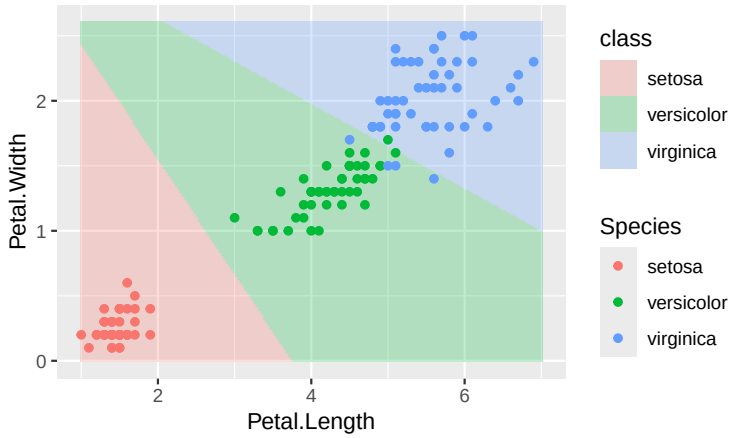
linear in x, for any number of classes.

QDA. Class-specific Σ_k: quadratic boundaries, more parameters. LDA often performs like logistic regression.

```
mod_lda <- lda(default ~ student + balance + income, data = train)
mod_qda <- qda(default ~ student + balance + income, data = train)
c(lda = accuracy(predict(mod_lda, test)$class, test$default),
  qda = accuracy(predict(mod_qda, test)$class, test$default))
```

```
##    lda    qda
## 0.9655 0.9680
```

```
lda2 <- lda(Species ~ Petal.Length + Petal.Width, data = iris)
grid <- expand_grid(Petal.Length = seq(1, 7, length = 200), Petal.Width = seq(0, 2.6,
    length = 200))
grid$class <- predict(lda2, grid)$class
ggplot(iris, aes(Petal.Length, Petal.Width)) +
  geom_raster(data = grid, aes(fill = class), alpha = 0.25) +
  geom_point(aes(colour = Species))
```



LDA boundaries on iris are straight lines.

8.4 Logistic regression for classification

Motivation. Least squares fails for logistic regression; cross-entropy, i.e. maximum likelihood, works.

$\text{logit}(q) = x^\top\beta$ gives $q = P(y = 1 \mid x) = \text{expit}(x^\top\beta) = 1/(1 + e^{-x^\top\beta})$; $q > 0.5$ iff $x^\top\beta > 0$, so the **decision boundary** is the hyperplane $x^\top\beta = 0$.

High earners. On capital gains alone the slides get $\hat{\beta}_0 = -1.37$, $\hat{\beta}_1 = 0.000335$: high-income iff gains exceed $-\hat{\beta}_0/\hat{\beta}_1 \approx \$4\,102$. On `Default` with balance:

```
mod_log_1 <- glm(default ~ balance, family = binomial, data = train)
tidy(mod_log_1)
```

```
## # A tibble: 2 x 5
##   term         estimate std.error statistic   p.value
##   <chr>           <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept) -10.8      0.421        -25.8 1.56e-146
## 2 balance       0.00559  0.000257      21.8 3.12e-105
```

```
-coef(mod_log_1)[1] / coef(mod_log_1)[2]
```

```
## (Intercept)
##     1939.69
```

Defaulter iff balance exceeds about \$1 940.

8.4.1 Why not squared loss

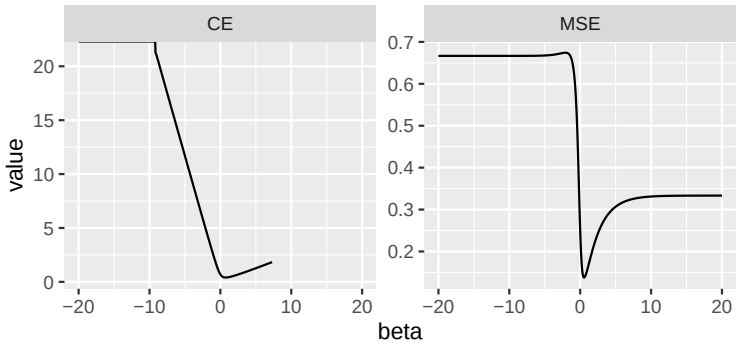
The MSE

$$\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \operatorname{expit}(x_i^\mathsf{T}\beta)\right)^2$$

is not convex in β. Toy data, $\operatorname{logit} q = \beta x$:

```
x <- c(-4, -2, -0.5, 1, 3, 5); y <- c(0, 0, 1, 0, 1, 1)
expit <- function(z) 1 / (1 + exp(-z))
tibble(beta = seq(-20, 20, by = 0.05)) %>%
  mutate(MSE = map_dbl(beta, ~ mean((y - expit(.x * x))^2)),
         CE = map_dbl(beta, ~ -mean(y * log(expit(.x * x)) + (1 - y) * log(1 - expit(.x
    * x))))) %>%
  pivot_longer(-beta) %>%
  ggplot(aes(beta, value)) + geom_line() + facet_wrap(~ name, scales = "free_y")
```

```
## Warning: Removed 254 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```
mse <- function(b) mean((y - expit(b * x))^2)
sapply(c(-20, -10.34, -2, 0.54, 5), mse)
```

```
## [1] 0.6666667 0.6666720 0.6741982 0.1378328 0.3067829
```

```
optimize(mse, c(-5, 5))$minimum
```

```
## [1] 0.5446891
```

The MSE (right) has a hump near $\beta = -2$ and a plateau at 2/3 for negative β: gradient descent from the left stalls (slides: $\beta = -10.34$), far from the minimum $\beta \approx 0.54$.

8.4.2 Cross-entropy loss

For binary y and predicted probability q, the **cross-entropy** (log) loss is

$$-\{y \log q + (1-y)\log(1-q)\}.$$

($-\log q$ if $y = 1$: infinite as $q \to 0$). It punishes confident errors and its average

$$-\frac{1}{n}\sum_{i=1}^{n}\left\{y_i \log \operatorname{expit}(x_i^\mathsf{T}\beta) + (1-y_i)\log\left(1 - \operatorname{expit}(x_i^\mathsf{T}\beta)\right)\right\},$$

is convex in β (left panel).

Convexity of cross-entropy [Supplement]

With $q_i = \operatorname{expit}(x_i^\top \beta)$, the gradient of the average loss is $\frac{1}{n}\sum_i (q_i - y_i)x_i$ and the Hessian is $\frac{1}{n}\sum_i q_i(1-q_i)x_i x_i^\top$, positive semidefinite; Newton's method (IRLS, as in `glm()`) finds the global minimum when it exists.

Connection to maximum likelihood. Given x_i, $y_i \sim \text{Bernoulli}(q_i)$ with likelihood $q_i^{y_i}(1-q_i)^{1-y_i}$, so the log-likelihood is

$$\sum_{i=1}^{n} \{y_i \log q_i + (1-y_i)\log(1-q_i)\},$$

$-n$ times the average cross-entropy: minimising cross-entropy *is* maximum likelihood (as least squares is for normal errors, ridge/lasso with a prior on β).

More than two classes. Chains of binary models or the multinomial model (`nnet::multinom()`); discriminant analysis is more common.

Summary. Naive Bayes: conditional independence; LDA/QDA: normal classes; logistic regression: linear logit $P(y = 1 \mid x)$ fitted by maximum likelihood.

8.5 Selected exercises

Motivation. Naive Bayes (binary predictors) and LDA give a linear log-odds, the logistic form; QDA does not.

Exercise 8.1 (naive Bayes is linear in the log-odds)

Two classes $y \in \{0, 1\}$ with priors π_0, π_1, and binary predictors $x_1, \ldots, x_p$ that are conditionally independent given y, with $p_{jk} = P(x_j = 1 \mid y = k) \in (0, 1)$. Show that

$$\log \frac{P(y=1 \mid x)}{P(y=0 \mid x)} = w_0 + \sum_{j=1}^{p} w_j x_j,$$

and give $w_0, \ldots, w_p$. How does the naive Bayes fit of these coefficients differ from a logistic regression fit?

Solution [Supplement].

Step 1. By Bayes' theorem the denominator cancels in the ratio, and by conditional independence $P(x \mid y = k) = \prod_j p_{jk}^{x_j}(1-p_{jk})^{1-x_j}$. Hence

$$\log \frac{P(y=1 \mid x)}{P(y=0 \mid x)} = \log \frac{\pi_1}{\pi_0} + \sum_j \Big\{ x_j \log \frac{p_{j1}}{p_{j0}} + (1-x_j)\log \frac{1-p_{j1}}{1-p_{j0}} \Big\}.$$

Step 2. Collecting the terms in x_j,

$$w_j = \log \frac{p_{j1}(1-p_{j0})}{p_{j0}(1-p_{j1})}, \qquad w_0 = \log \frac{\pi_1}{\pi_0} + \sum_j \log \frac{1-p_{j1}}{1-p_{j0}};$$

w_j is the log odds ratio between x_j and y.

Step 3. Same family of posteriors. Naive Bayes estimates each w_j from class-wise frequencies of x_j alone; logistic regression fits all w_j jointly by maximum likelihood of y given x, adjusting for correlation (naive Bayes double-counts correlated evidence). ■

Exercise 8.2 (LDA implies a logistic model)

Let $x \mid y = k \sim N(\mu_k, \Sigma)$ for $k = 0, 1$, with priors π_0, π_1 and Σ positive definite. Show that $P(y = 1 \mid x) = \operatorname{expit}(\beta_0 + \beta^\top x)$ with $\beta = \Sigma^{-1}(\mu_1 - \mu_0)$, and find β_0. Show that for $\pi_0 = \pi_1$ the decision boundary passes through $(\mu_0 + \mu_1)/2$.

Solution [Supplement].

Step 1. The normalising constants of the two densities are equal (same Σ), so

$$\log \frac{\pi_1 f_1(x)}{\pi_0 f_0(x)} = \log \frac{\pi_1}{\pi_0} - \frac{1}{2}(x - \mu_1)^\top \Sigma^{-1}(x - \mu_1) + \frac{1}{2}(x - \mu_0)^\top \Sigma^{-1}(x - \mu_0).$$

Step 2. The quadratic terms $x^\top \Sigma^{-1} x$ cancel, and by symmetry of Σ^{-1} the cross terms give $x^\top \Sigma^{-1}(\mu_1 - \mu_0)$. So the log-odds equals $\beta_0 + \beta^\top x$ with

$$\beta = \Sigma^{-1}(\mu_1 - \mu_0), \qquad \beta_0 = \log \frac{\pi_1}{\pi_0} - \frac{1}{2}(\mu_1^\top \Sigma^{-1} \mu_1 - \mu_0^\top \Sigma^{-1} \mu_0),$$

and $P(y = 1 \mid x) = \operatorname{expit}(\beta_0 + \beta^\top x)$.

Step 3. For $\pi_0 = \pi_1$ and $m = (\mu_0 + \mu_1)/2$, $\beta^\top m = \frac{1}{2}(\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 + \mu_0) = \frac{1}{2}(\mu_1^\top \Sigma^{-1} \mu_1 - \mu_0^\top \Sigma^{-1} \mu_0) = -\beta_0$, because the mixed terms $\mu_1^\top \Sigma^{-1} \mu_0$ and $\mu_0^\top \Sigma^{-1} \mu_1$ are equal. So m lies on the boundary $\beta_0 + \beta^\top x = 0$, which generalises Proposition 8.1.

Step 4. The converse fails (non-normal classes can give linear log-odds): logistic regression is more robust, LDA more efficient under normality. ∎

Exercise 8.3 (QDA boundaries can be two points)

Let $x \mid y = k \sim N(\mu_k, \sigma_k^2)$ for $k = 0, 1$, with equal priors.
(a) Show that the set where the Bayes rule chooses class 1 is defined by a quadratic inequality in x when $\sigma_0 \neq \sigma_1$.
(b) For $\mu_0 = \mu_1 = 0$, $\sigma_0 = 1$, $\sigma_1 = 2$, find the rule. What does LDA do with these data?

Solution [Supplement].

(a) Class 1 is chosen when $\log f_1(x) > \log f_0(x)$, i.e.

$$-\log \sigma_1 - \frac{(x - \mu_1)^2}{2\sigma_1^2} > -\log \sigma_0 - \frac{(x - \mu_0)^2}{2\sigma_0^2}.$$

The coefficient of x^2 in the difference is $\frac{1}{2\sigma_0^2} - \frac{1}{2\sigma_1^2} \neq 0$, so this is a quadratic inequality; its solution set is an interval or the complement of an interval (possibly empty or all of $\mathbb{R}$), with at most two boundary points.

(b) $-\log 2 - x^2/8 > -x^2/2$, i.e. $\frac{3}{8}x^2 > \log 2$, i.e. $|x| > \sqrt{8 \log 2/3} \approx 1.36$. LDA pools variances; with $\mu_0 = \mu_1$ its discriminants differ only by $\log \hat{\pi}_k$, so it assigns every point to one class. ∎